Lesson 4: Sum to Infinity & Applications — Practice Worksheet

Name: ____________________________ Date: ___________

Attempt every question. Show your working. The sections increase in difficulty.

Section 1 — Warm-up (foundation)

1. Find the sum to infinity of 12 + 6 + 3 + …

2. Find the sum to infinity of 27 + 9 + 3 + …

3. Does 3 + 6 + 12 + … converge? Explain.

4. Find S_∞ of 100 + 20 + 4 + …

5. State the condition on r for a geometric series to have a sum to infinity.

6. Find S_∞ of 50 + 25 + 12.5 + …

Section 2 — Core practice

7. A geometric series has $u_1 = 40$ and $r = 0.25$. Find the sum to infinity.

8. Find S_∞ of $6 - 3 + 1.5 - \dots$

9. A series has $u_1 = 60$ and $S_\infty = 80$. Find r.

10. A series has $r = 0.6$ and $S_\infty = 50$. Find u_1.

11. Find S_∞ of 81 + 27 + 9 + …

Section 3 — Challenge & exam-style

12. Write the recurring decimal 0.7 as a geometric series and hence as a fraction.

13. Write 0.27 (i.e. 0.272727…) as a fraction using $S\infty$.

14. A ball is dropped from 10 m and each bounce reaches 0.6 of the previous height. Find the total vertical distance travelled.

15. A geometric series has first term 18 and sum to infinity 30. Find r, then the sum of the first 3 terms.